pfi

# Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

# Market Data Tools

WEEK 2 · DAY 1

Created by **Jeff Lenamon**

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# Today: market data tools

## Every question names a date

**2,940** days of the curve, from the repo

**14** tenors, the Treasury's labels

**5** tools, curve and spreads

## Today

1. market_tools over two files
2. A date the file lacks
3. Changes in bp and slopes, checked
4. Four questions over the curve
5. FRED as a tool, live only
6. The licensed series, refused
7. Synthetic, named synthetic

**No tool reads the clock. A missing date is an error result. Recorded runs read files only.**

# The rules for agents

## The curve is described, never forecast

### Allowed names

the 10 Yr par yield in the file

the change in bp between two dates

the slope in bp

### Never called

a forecast or an outlook

a signal

a view on rates

## What an agent produces: text, every number from a tool

### Allowed names

the agent returned

the saved run

the brief states

### Never called

the agent's opinion or view

what the agent believes

the agent's advice

# market_tools: five tools over two files

### market_tools(curve, spreads)

| | |
|---|---|
| `treasury_yield` | one tenor, one date: yield_pct |
| `curve_on` | every tenor, one date: yield_pct rows |
| `curve_change_bp` | one tenor, two dates: change_bp |
| `slope_bp` | long less short, one date: slope_bp |
| `synthetic_spread` | IG or HY, one date: spread_bp |

### Why files

A tool result goes into the next request

The request's key is its body

A live source changes the result,

so the key, so the replay stops

A committed file returns the same

number on any machine, any year

**The frames never leave the closure: the model sees only what a tool returns.**

# A date the file lacks

```
{'error': 'no row for that date in the curve file: 2026-10-03 10 Yr'}
asof 2026-10-03: 5.28 percent, the row of 2026-10-02
```

### The agent, in the saved run: treasury_yield, then the answer

"The 10 Yr Treasury par yield on 2026-10-03 is not available in the curve file."

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,064 prompt and 74 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

**The tool answers for the date asked, or says it cannot. Never the nearest row.**

# Changes in bp and slopes, checked

```
{'tenor': '10 Yr', 'start': '2026-09-02', 'end': '2026-10-02', 'start_yield_pct': 4.79, 'end_yiel...
pandas: 49.00000000000002 bp; marketdata.change_since_bp: 49.00000000000002 bp
{'short': '2 Yr', 'long': '10 Yr', 'as_of': '2026-10-02', 'slope_bp': 45.0}
marketdata.slope_bp: 45.000000000000014 bp
```

## x 100

a difference of two yields in percent, in bp

## same

pandas, change_since_bp, slope_bp: one arithmetic

## 6 decimals

every tool result: the float's tail rounded away

**The tool gets checked before any agent uses it.**

# Four questions over the curve

### In the saved runs: tools requested, then both checks

```
level   treasury_yield      2 requests  figures 1  passed 1  key 1 of 1
change  curve_change_bp     2 requests  figures 1  passed 1  key 1 of 1
slope   slope_bp            2 requests  figures 1  passed 1  key 1 of 1
curve   treasury_yield x 4  5 requests  figures 4  passed 4  key 4 of 4
```

### The slope run's first reply, written beside its tool request, before the tool returned

```
"The 2s10s slope of the Treasury curve on 2026-10-02, which is the 10 Yr par yield less the 2 Yr par yield, is 44 ba..."
```

numbers untraced in it: ['44']; the final answer cites the tool's result

**check_figures: the number came from the result it cites. The key: the right question was asked.**

# FRED as a tool

### fred_latest(series_id, as_of), in this order

1. A licensed prefix: an error result, nothing sent
2. cached_fred_series: once a day, in cache/fred/
3. The date asked, or an error result

### Live cells only, with your FRED key

Never in a recorded run

The build had no key, no network

Its saved output:

```
Live cell skipped: no FRED key was found....
```

This product uses the FREDÂ® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.

**A FRED value can be revised; the committed file cannot. FRED is for your own live cells.**

# The licensed series

```
{'error': "BAMLH0A0HYM2 is licensed data (ICE BofA or Moody's): this tool never downloads it"}
{'error': "BAA10Y is licensed data (ICE BofA or Moody's): this tool never downloads it"}
files in the FRED cache folder: []
```

**0**

values returned

**0**

files in the FRED cache

**None**

the key the tool was given: a refusal sends nothing

**Shown by asking the tool directly. A recorded run never holds a licensed series ID.**

# The synthetic spreads, named synthetic

**In the saved run: synthetic_spread, then synthetic_spread**

```
On 2026-10-02, the course's synthetic investment grade (IG) credit spread was 97.0 basis points,

... and the synthetic high yield (HY) credit spread was 328.7 basis points.
```

**the answer says synthetic: yes**

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 3,257 prompt and 212 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

**Invented by the course: no index, no issuer, no real spread series. The word travels with the number.**

# Excel: INDEX and MATCH, and #N/A

```
=INDEX($B$2:$O$2941,MATCH(DATE(2026,10,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0))
  returned 5.28
=(INDEX($B$2:$O$2941,MATCH(DATE(2026,10,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0))
  -INDEX($B$2:$O$2941,MATCH(DATE(2026,9,2),$A$2:$A$2941,0),MATCH("10 Yr",$B$1:$O$1,0)))*100   returned 49.00000000000002
```

**Typed into Excel 16.0 build 20430: 8 of 8 rows the same as the tools**

#N/A: 10 Yr on 2026-10-03; 10 Yr change, 2026-09-02 to 2026-10-03; 10 Yr on 2014-06-02

MATCH with 1, or XLOOKUP with -1, on the Saturday: 5.28, the Friday's row

**git log --oneline --stat -3: 3 commits, each with its files. 34 passed.**

# The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

## Docstring

Refuse a licensed prefix before anything is downloaded or written.

→

2

## Code back

Written for the course in the style of an AI answer. Read the order.

→

3

## Test first

A fake download records whether it was reached. Run it: it fails.

→

4

## Fix

Refuse first, then Course 3's cached_fred_series.

**Today: the refusal comes after the download and the write, so the licensed series lands in the cache.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Market data tools over a committed file
- A missing date is an error result
- Changes and slopes checked against pandas
- Figures checked twice: result and key
- FRED live only; licensed refused first

**Week 2, Day 2** Document tools: statements, indentures, 8-K sections